

Course Code	Name of the Course				
216U03C402	Analog Electronics				
Teaching Scheme	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	-	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		20	30	50	100

Course pre-requisites:

- Elements of Electrical and Electronics Engineering
- Basic Electronic Circuits

Course Objectives:

The objective of this course is to explore analysis and design of Analog Electronic Circuits-multistage amplifier, differential amplifier, operational amplifiers, oscillators and power amplifiers.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Analyse the frequency response of BJT and FET amplifier.
CO2. Understand the basics of Analog Integrated Circuits
CO3. Understand working and design of BJT power amplifiers
CO4. Explore oscillator circuits using operational amplifiers

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Frequency Response of BJT and MOSFET Amplifiers		16	CO1
	1.1	High frequency hybrid-pi model of BJT and MOSFET, Miller effect and capacitance and gain bandwidth product		
	1.2	Effect of coupling, bypass and junction capacitors on frequency response, low and high frequency response of single stage BJT - CE, CB, CC and MOSFET - CS, CG, CD amplifiers		
	1.3	Mid-frequency analysis of multistage - CS-CS and Cascode CS- CG amplifiers and Darlington pair, design of two stage amplifiers		
	1.4	Low and high frequency response of multistage - CS-CS and Cascode CS-CG amplifiers		
2	Biasing Techniques for Integrated Circuits		06	CO2
	2.1	BJT-MOSFET two and three transistor current sources		
	2.2	Wilson and Widlar current sources		
3	Fundamentals of Operational Amplifier		09	CO2
	3.1	Practical Op-Amp parameters, Finite open loop gain, frequency response, offset voltage, input bias current, CMRR		
	3.2	Op-amp with negative feedback, block diagram representation of feedback Configurations: Voltage-Series, Voltage-shunt, Current-Series, Current-shunt. Analysis of Voltage-Series Feedback Amplifier		
	3.3	Analysis and design of Adder, Subtractor, Active filters, V to I converters, Schmitt Trigger, Voltage Comparators, Logarithmic amplifiers, Sample and Hold circuit, Peak detector		
4	BJT Power amplifier		09	CO3
	4.1	Amplifier classes, Series-Fed Class A, Transformer coupled Class A amplifier, Class B, Class C, Class D, Class AB amplifiers		
	4.2	DC analysis, Power efficiency, Distortion and Heat Sink		
	4.3	Design of class A power amplifier		
5	Oscillator Circuits		05	CO4
	5.1	Concept of positive feedback, conditions for oscillations and Barkhausen Criteria for sustained oscillations		
	5.2	Hartley oscillator, Crystal oscillator, Phase shift oscillator, Wien Bridge oscillator		

	Total	45	
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Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/ Year
1.	Donald A. Neamen	<i>Electronic Circuit Analysis and Design</i>	Tata McGraw Hill, India	3rd Edition 2014
2.	Adel S. Sedra, Kenneth C. Smith, Arun N Chandorkar	<i>Microelectronic Circuits Theory and Applications</i>	OXFORD International Edition, India	6th Edition, 2013
3.	Robert Boylestad , Louis Nashelsky	<i>Electronic Devices and Circuit Theory</i>	Pearson Education, India	10th Edition 2015
4.	Sergio Franco	<i>Design with Operational Amplifiers and analog integrated circuits</i>	McGraw Hill International, New Delhi, India	3rd Edition 2017
5.	Ramakant A. Gayakwad	<i>Op-Amps and Linear Integrated Circuit</i>	Pearson Prentice Hall, India	4th Edition, 2012